

BCS 209: PRINCIPLE OF NETWORKS

1. Introduction: Definitions

A network is a collection of computers (nodes) and transmission channels (links) that allow **people** to communicate over distances, large and small. A Bluetooth personal area network may simply connect a home PC with its peripherals. An undersea fiber optic cable may traverse an ocean. The Internet and telephone networks span the globe.

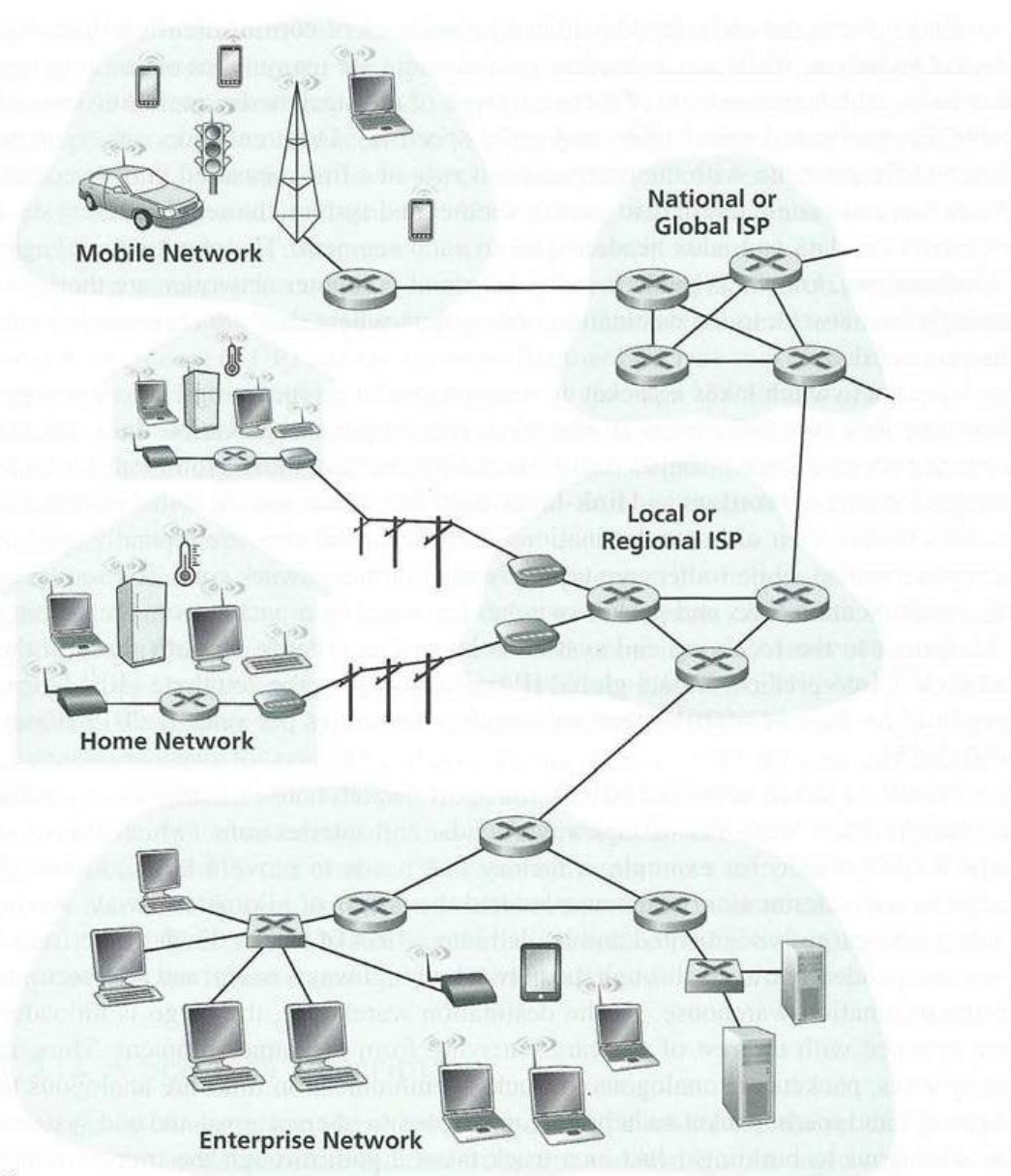
Computer network: A system of interconnected computers and computerized peripherals such as printers. This interconnection among **computers** facilitates information sharing among them. Computers may connect to each other by either wired or wireless media.

Internetworking: It is a network of networks or simply the internet. It is the largest network in existence on this planet and consists of hundreds of millions of connected computers, communication links, and switches; with billions of users who connect via laptops, tablets, and smartphones; and with an array of new Internet-connected "things" including game consoles, surveillance systems, watches, eye glasses, thermostats, body scales, and cars.

The internet hugely connects all Wide Area Networks (WANs) and it can have connection to Local Area Networks (LANs) and Home networks. Internet uses TCP/IP protocol suite and uses IP as its addressing protocol. Currently, Internet is widely implemented using IPv4. Because of shortage of address spaces, it is gradually migrating from IPv4 to IPv6.

By some estimates, in 2015 there were about 5 billion devices connected to the Internet, and the number will reach 25 billion by 2020 [Gartner 2014]. It was estimated that in 2015 there were over 3.2 billion Internet users worldwide.

Some graphical representation of the internet



Key:

- | | | | | | | | | |
|---|---|---|---|---|--|---|---|---|
|  |  |  |  |  |  |  |  |  |
| Host
(= end system) | Server | Mobile | Router | Link-layer
switch | Modem | Base
station | Smartphone | Cell phone
tower |
|  |  |  |  |  |  | | | |
| Tablet | Traffic light | Thermostat | Fridge | Flat computer
monitor | Keyboard | | | |

Administration:

From an administrator's view point, a network can be private, which belongs a single autonomous system and cannot be accessed outside its physical or logical domain. A network can be public, which is accessed by all.

Network Architecture

Computer networks are classified into various types such as Client-Server, peer-to-peer or hybrid, depending upon its architecture.

- i. Where one or more systems acting as Server and the other being client, requests the Server to serve requests. Server takes and processes request on behalf of Clients.
- ii. Two systems can be connected Point-to-Point, or in back-to-back fashion. They both reside at the same level and called peers.
- iii. There can be hybrid network which involves network architecture of both the above types.

Computer Network Applications

The major reason why computer networks exist or are created is to allow for network applications to run. The following are some of the advantages of the networks:

- i. Resource sharing such as printers and storage devices
- ii. Exchange of information by means of e-Mails and FTP
- iii. Information sharing by using Web or Internet
- iv. Interaction with other users using dynamic web pages
- v. IP phones
- vi. Video conferences
- vii. Parallel computing
- viii. Instant messaging

DATA TRANSMISSION

Data or information transmission is the movement of data from one node/end to the other for the purposes of communication. This transmission can be said to be in either digital or analog mode.

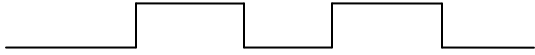
Signals

When data is sent over physical medium, it has to be first converted into electromagnetic signals. Data itself can either be analog like human voice, or

digital for example file on the disk. Both analog and digital data are represented in digital or analog signals.

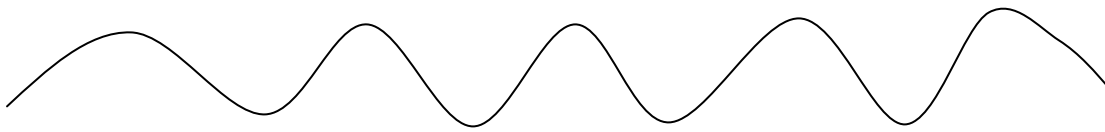
Digital Signals

Digital signals are discrete in nature and represent sequence of voltage pulses. Digital signals are used within the circuitry of a computer system.



Analog Signals

Analog signals are in continuous wave form in nature and represented by continuous electromagnetic waves.



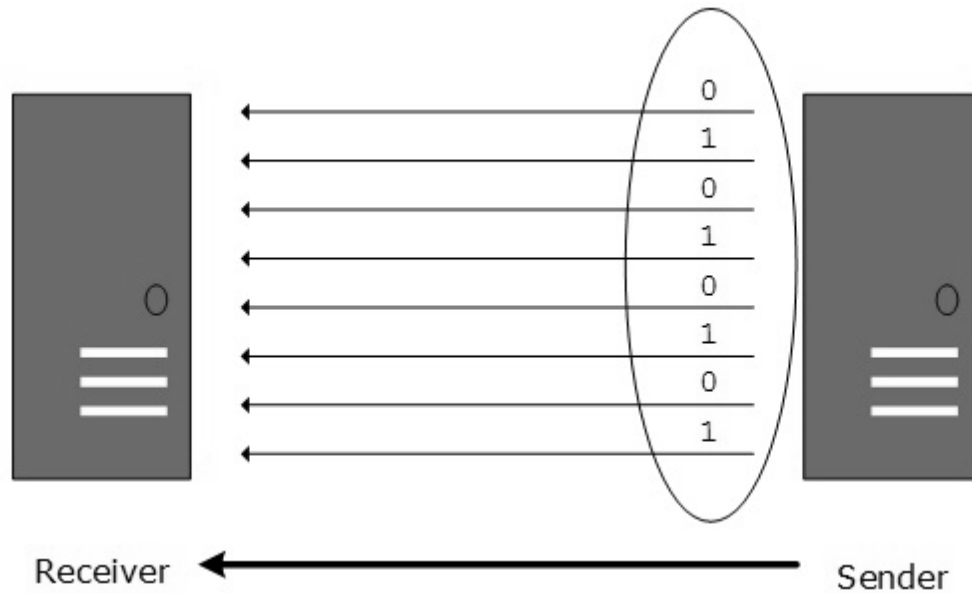
DIGITAL TRASMISSION

For a computer to use the data, it must be in discrete digital form. Similar to data, signals can also be in analog and digital form. To transmit data digitally, it needs to be first converted to digital form.

Transmission Modes

The transmission mode decides how data is transmitted between two computers. The binary data in the form of 1s and 0s can be sent in two different modes: Parallel and Serial.

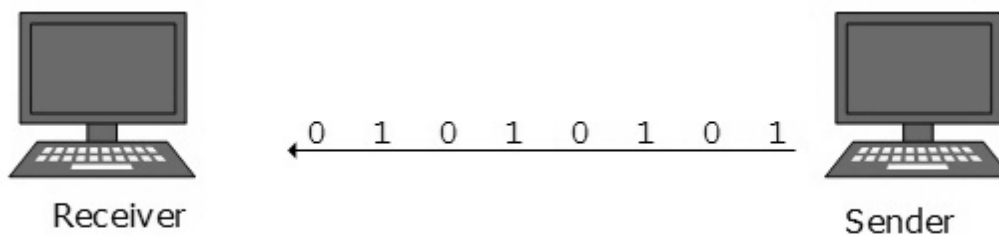
Parallel Transmission



The binary bits are organized into groups of fixed length. Both sender and receiver are connected in parallel with the equal number of data lines. Both computers distinguish between high order and low order data lines. The sender sends all the bits at once on all lines. Because the data lines are equal to the number of bits in a group or data frame, a complete group of bits (data frame) is sent in one go. Advantage of Parallel transmission is high speed and disadvantage is the cost of wires, as it is equal to the number of bits sent in parallel.

Serial Transmission

In serial transmission, bits are sent one after another in a queue manner. Serial transmission requires only one communication channel.



Serial transmission can be either asynchronous or synchronous.

Asynchronous Serial Transmission

This is where there is no significance given to timing. Data-bits have specific pattern and they help receiver recognize the start and end data bits. For example, a 0 is prefixed on every data byte and one or more 1s are added at the end.

Two continuous data-frames (bytes) may have a gap between them.

Synchronous Serial Transmission

Timing in synchronous transmission has importance as there is no mechanism followed to recognize start and end data bits. There is no pattern or prefix/suffix method. Data bits are sent in burst mode without maintaining gap between bytes (8-bits). Single burst of data bits may contain a number of bytes. In this case, timing becomes very important.

It is up to the receiver to recognize and separate bits into bytes. The advantage of synchronous transmission is high speed, and it has no overhead of extra header and footer bits as in asynchronous transmission.

Transmission Media

The media over which the information between two computer systems is sent is called transmission media. Transmission media comes in two forms:

Guided Media

All communication wires/cables are guided media, such as **Unshielded Twisted Pair (UTP) Cables, coaxial cables, and fiber Optics**. In this media, the sender and receiver are directly connected and the information is sent (guided) through it.

1. Unshielded Twisted Pair (UTP) Cable



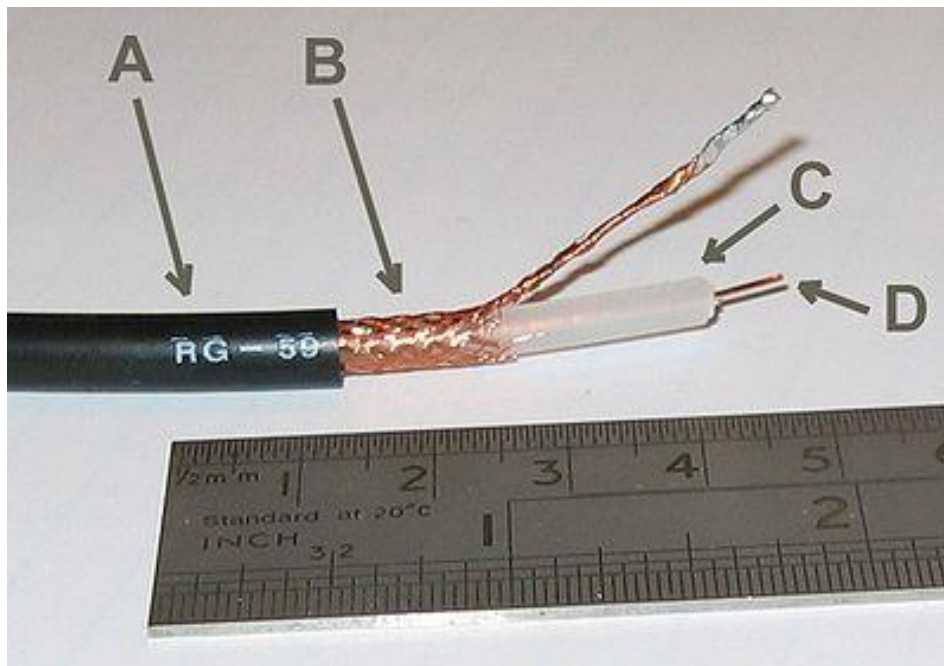
Various cables are used for carrying Ethernet: current common types include Cat 5, 5e, Cat 6, 6a, 7 and Cat 8 and the RJ45 connector is widely used.

2. Coaxial Cable

Coaxial cable has two wires of copper. The core wire lies in the center and it is made of solid conductor. The core is enclosed in an insulating sheath. The second wire is wrapped around over the sheath and that too in turn encased by insulator sheath. This all is covered by plastic cover.

Because of its structure, the coax cable is capable of carrying high frequency signals than that of twisted pair cable. The wrapped structure provides it a good shield against noise and cross talk. Coaxial cables provide high bandwidth rates of up to 450 mbps.

Cables are connected using BNC connector and BNC-T. BNC terminator is used to terminate the wire at the far ends.



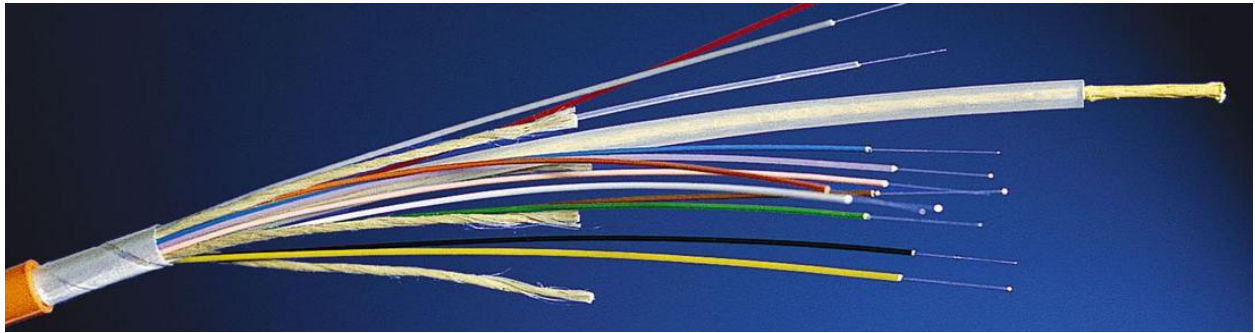
- A. Outer plastic sheath
- B. Woven copper shield
- C. Inner dielectric insulator
- D. Copper core

3. Fiber Optic Cable:

Fiber Optic operates on the properties of light. When light ray hits at critical angle, it tends to refract at 90 degree. This property has been used in fiber optic. The core of fiber optic cable is made of high quality glass or plastic. From one end of it light is emitted, it travels through it and

at the other end light detector detects light stream and converts it to electric data.

Fiber Optic provides the highest mode of speed. It comes in two modes, one is single mode fiber and second is multimode fiber. Single mode fiber can carry a single ray of light whereas multimode is capable of carrying multiple beams of light.



Unguided Media

Wireless or open air space is said to be unguided media, since there is no physical connectivity between the sender and receiver. Information is spread over the air, and anyone including the actual recipient may collect the information. Unguided technology includes **microwave line of sight, satellites, cellular systems, ad hoc networks and wireless sensor networks.**

Channel Capacity

The speed of transmission of information is said to be the channel capacity. We count it as data rate in digital world. It depends on numerous factors such as:

- i. **Bandwidth:** The physical limitation of underlying media.
- ii. **Error-rate:** Incorrect reception of information because of noise.
- iii. **Encoding:** The number of levels used for signaling.

Transmission Impairment

When signals are traversing or are being transmitted over a medium, they tend to deteriorate. This may be as a result of the following reasons:

Dispersion

As signal travels through the media, it tends to spread and overlaps. The amount of dispersion depends upon the frequency used.

Attenuation

For the receiver to interpret the data accurately, the signal must be sufficiently strong. When the signal passes through the medium, it tends to get weaker. As it covers distance, it loses strength. This situation is called **attenuation**.

Delay distortion

Signals are sent over media with pre-defined speed and frequency. If the signal speed and frequency do not match, there are possibilities that signal reaches destination in an arbitrary fashion. In digital media, this is very critical that some bits reach earlier than the previously sent ones.

Noise

Random disturbance or fluctuation in analog or digital signal is said to be Noise in signal, which may distort the actual information being carried. Noise can be characterized in one of the following class:

i. Thermal Noise

Heat agitates the electronic conductors of a medium which may introduce noise in the media. Up to a certain level, thermal noise is unavoidable.

ii. Intermodulation

When multiple frequencies share a medium, their interference can cause noise in the medium. Intermodulation noise occurs if two different frequencies are sharing a medium and one of them has excessive strength or the component itself is not functioning properly, then the resultant frequency may not be delivered as expected.

iii. Crosstalk

This sort of noise happens when a foreign signal enters into the media. This is because signal in one medium affects the signal of second medium.

iv. Impulse

This noise is introduced because of irregular disturbances such as lightening, electricity, short-circuit, or faulty components. Digital data is mostly affected by this sort of noise.

TYPES OF COMPUTER NETWORKS

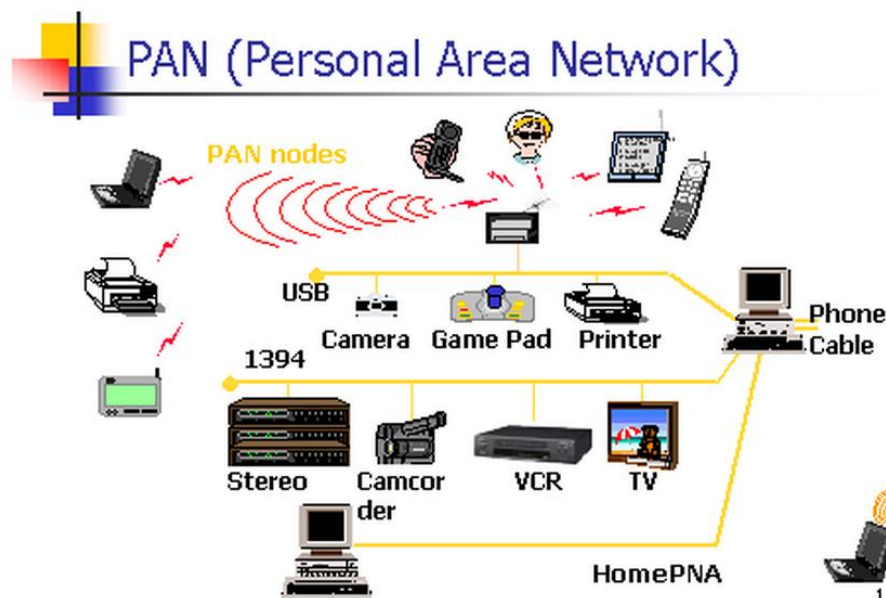
Networks are distinguished based on their geographical span. A network can be as small as distance between your mobile phone and its Bluetooth headphone and as large as the internet itself, covering the whole geographical world.

The following are various types of networks:

1. Personal Area Network (PAN)
2. Local Area Network (LAN)
3. Metropolitan Area Network
4. Wide Area Network (WAN)
5. Internetwork/Internet

A. Personal Area Network (PAN)

It is the smallest network that is very personal to a user. It covers connectivity range of up to 10 meters, is low power and of low cost. PAN may include **Bluetooth** or **infra-red** enabled devices, **wireless computer keyboard** and **mouse**, Bluetooth enabled headphones, **wireless printers**, and **TV remotes**.



B. Local Area Network (LAN)

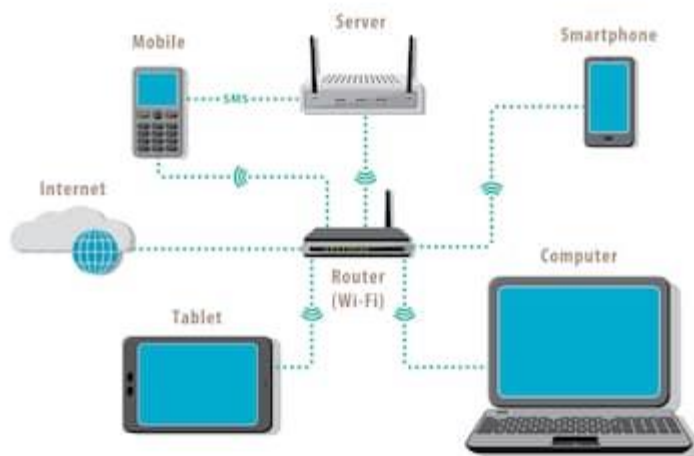
This is a network that spans inside a building and operated under single administrative system. Usually, LAN covers an organization offices, schools,

colleges or universities. Number of systems connected in LAN may vary from as least as two to some millions.

LAN provides a very useful way sharing the resources such as printers, file servers, scanners, and internet between end users. These resources are easily sharable among computers/devices.

LANs may contain local servers serving file storage and other locally shared applications. It mostly operates on private IP addresses and does not involve heavy routing. LAN works under its own local domain and controlled centrally.

LAN can be wired, wireless, or in both forms at once.

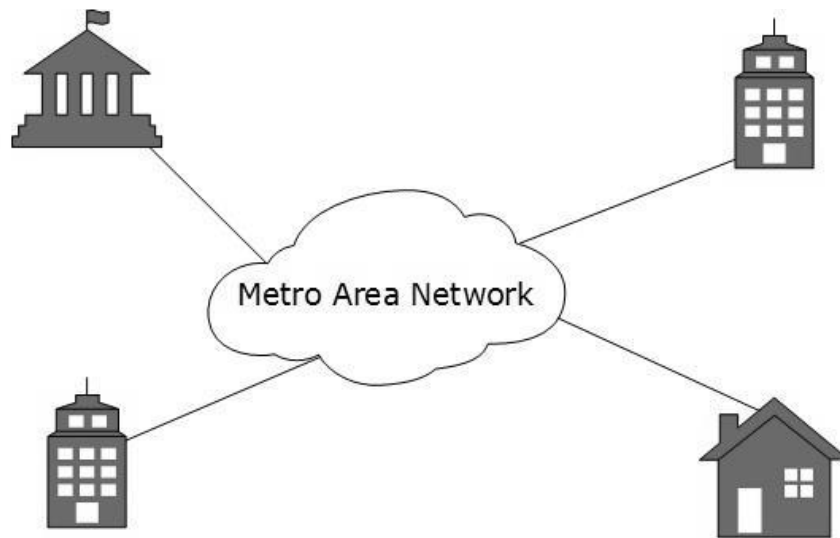


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C. Metropolitan Area Network(MAN)

The Metropolitan Area Network (MAN) generally expands throughout a city such as cable TV network. It can be in the form of Ethernet, Token-ring, ATM, or Fiber Distributed Data Interface (FDDI).

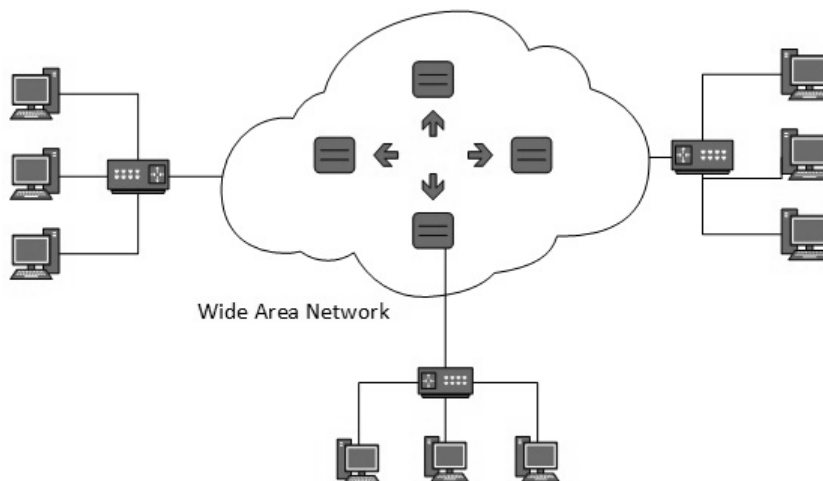
Metro Ethernet is a service which is provided by ISPs. This service enables its users to expand their Local Area Networks. For example, MAN can help an organization to connect all of its offices in a city.



Backbone of MAN is high-capacity and high-speed fiber optics. MAN works in between Local Area Network and Wide Area Network. MAN provides uplink for LANs to WANs or internet.

D. Wide Area Network (WAN)

The Wide Area Network (WAN) covers a wide area which may span across provinces and even a whole country. Generally, telecommunication networks are Wide Area Network. These networks provide connectivity to MANs and LANs. Since they are equipped with very high speed backbone, WANs use very expensive network equipment.



WAN use advanced technologies such as Asynchronous Transfer Mode (ATM), Frame Relay, and Synchronous Optical Network (SONET) and may be managed by multiple administrations.

E. Internetwork/Internet

This is a network of networks. It is the largest network in existence on this planet. The internet hugely connects all WANs and it can have connection to LANs and Home networks.

Internet uses TCP/IP protocol suite and uses IP as its addressing protocol. Present day, Internet is widely implemented using IPv4. Because of shortage of address spaces, it is gradually migrating from IPv4 to IPv6.

Internet enables its users to share and access enormous amount of information worldwide. It uses WWW, FTP, email services, audio, and video streaming etc. At huge level, internet works on Client-Server model.

Internet uses very high speed backbone of fiber optics. To inter-connect various continents, fibers are laid under sea known to us as submarine communication cable.

Internet is widely deployed on World Wide Web services using HTML linked pages and is accessible by client software known as Web Browsers like internet explorer, Mozilla/Firefox, Google chrome, Opera, Safari, Microsoft Edge, Maxthon, Flock, Seamonkey, Avant Browser, Netscape, Amaya etc....

When a user requests a page using some web browser located on some Web Server anywhere in the world, the Web Server responds with the proper HTML page. The communication delay is very low.

Internet is serving many proposes and is involved in many aspects of life. Some of them are:

- ✓ Web sites
- ✓ Marketing
- ✓ Networking
- ✓ Resource Sharing
- ✓ E-mail
- ✓ Instant Messaging
- ✓ Blogging
- ✓ Social Media

- ✓ Audio and Video Streaming

NETWORK LAN TECHNOLOGIES

Ethernet

This is a widely deployed wired LAN Technology and was invented by Bob Metcalfe and D.R. Boggs in the year 1970. It was standardized in IEEE 802.3 in 1980.

Ethernet shares media. Network which uses shared media has high probability of data collision. Ethernet uses Carrier Sense Multi Access/Collision Detection (CSMA/CD) technology to detect collisions. On the occurrence of collision in Ethernet, all its hosts roll back, wait for some random amount of time, and then re-transmit the data.

Ethernet connector is network interface card(NIC) equipped with 48-bits Media Access Control (MAC)address. This helps other Ethernet devices to identify and communicate with remote devices in Ethernet.

Ethernet follows Star topology with segment length up to 100 meters. All devices are connected to a hub/switch in a star fashion.

Fast-Ethernet

To encompass need of fast emerging software and hardware technologies, Ethernet extended itself as Fast-Ethernet. It can run on UTP, Optical Fiber, and wirelessly too and can provide speed of up to 100MBPS. This standard is named as 100BASE-T in IEEE 803.2 using Cat-5 twisted pair cable.

It uses CSMA/CD technique for wired media sharing among the Ethernet hosts and CSMA/CA (CA stands for Collision Avoidance) technique for wireless Ethernet LAN.

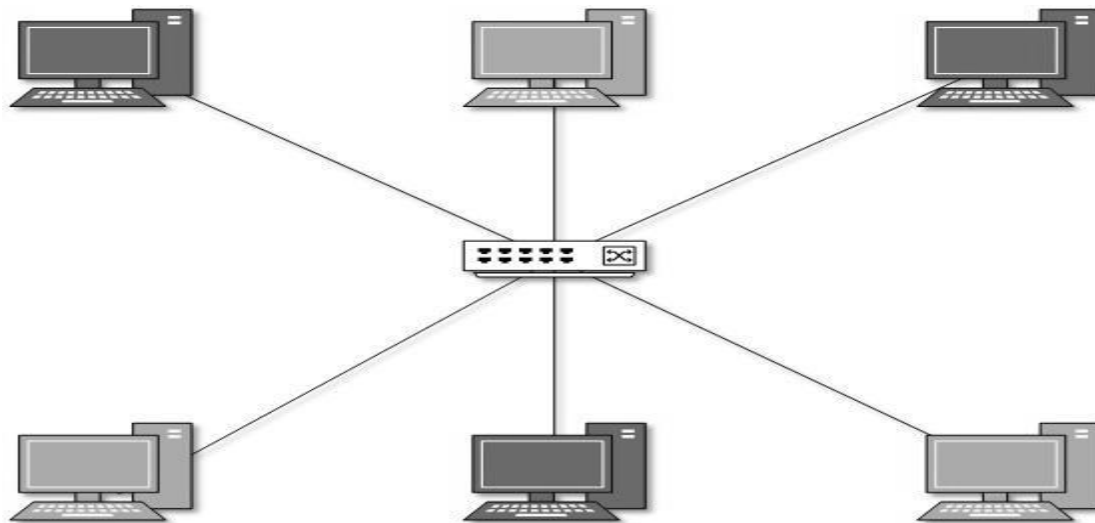
Fast Ethernet on fiber is defined under 100BASE-FX standard which provides speed up to 100MBPS on fiber. Ethernet over fiber can be extended up to 100 meters in half-duplex mode and can reach maximum of 2000 meters in full-duplex over multimode fibers.

Giga-Ethernet

After being introduced in 1995, Fast-Ethernet retained its high speed status only for three years till Giga-Ethernet was introduced. Giga-Ethernet provides speed up to 1000 mbits/seconds. IEEE802.3ab standardizes Giga-Ethernet over UTP using Cat-5, Cat-5e and Cat-6 cables. IEEE802.3ah defines Giga-Ethernet over Fiber.

Virtual LAN (VLAN)

LAN uses Ethernet which in turn works on shared media. Shared media in Ethernet create one single Broadcast domain and one single Collision domain. Introduction of switches to Ethernet has removed single collision domain issue and each device connected to switch works in its separate collision domain. But even Switches cannot divide a network into separate Broadcast domains. Virtual LAN is a solution to divide a single Broadcast domain into multiple Broadcast domains. Host in one VLAN cannot speak to a host in another. By default, all hosts are placed into the same VLAN.



Different VLANs are depicted in different color codes. Hosts in one VLAN, even if connected on the same Switch cannot see or speak to other hosts in different VLANs. VLAN is Layer-2 technology which works closely on Ethernet. To route

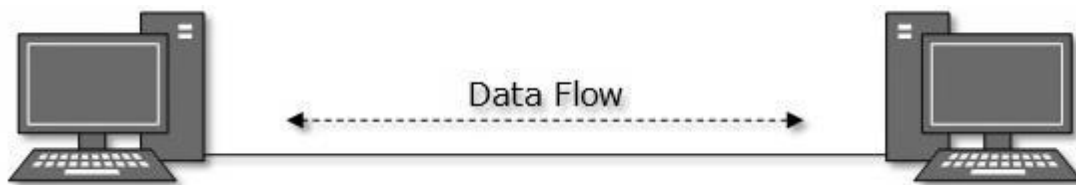
packets between two different VLANs, a Layer-3 device such as Router is required.

COMPUTER NETWORK TOPOLOGIES

This is the arrangement with which computer systems or network devices are connected to each other or one another. Topologies may define both physical and logical aspect of the network. Both logical and physical topologies could be same or different in a same network.

POINT-TO-POINT

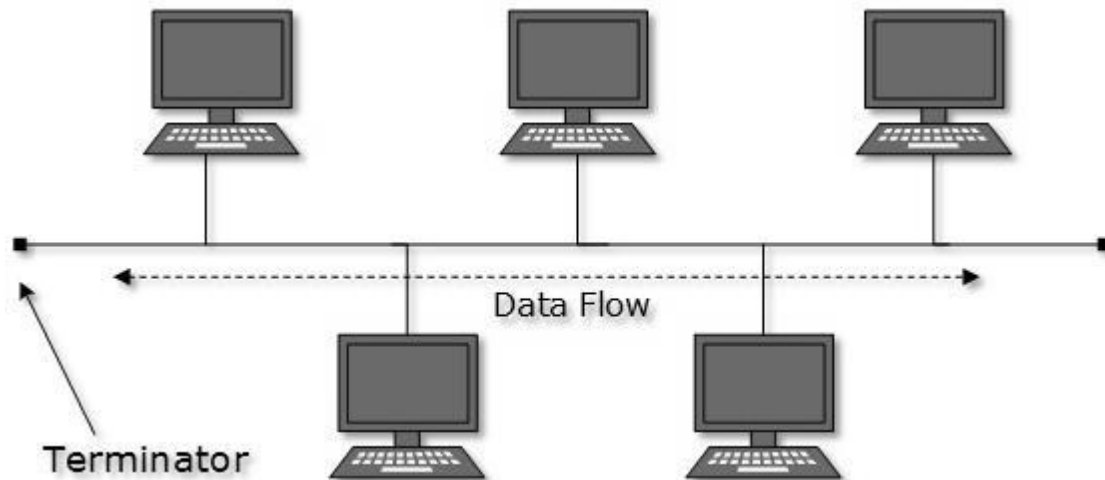
Point-to-point networks contain exactly two hosts such as computer, switches, routers, or servers connected back to back using a single piece of cable. Often, the receiving end of one host is connected to sending end of the other and vice versa.



If the hosts are connected point-to-point logically, then may have multiple intermediate devices. But the end hosts are unaware of underlying network and see each other as if they are connected directly.

Bus Topology

In this topology, all devices share single communication line or cable. This topology could experience problems while multiple hosts sending data at the same time. Therefore, Bus topology either uses CSMA/CD technology or recognizes one host as Bus Master to solve the issue. It is one of the simple forms of networking where a failure of a device does not affect the other devices. But failure of the shared communication line could make all other devices stop functioning.



Both ends of the shared channel have line terminator. The data is sent in only one direction and as soon as it reaches the extreme end, the terminator removes the data from the line.

Advantages:

- ✓ Ease of installation.
- ✓ Uses less cabling than mesh or star topologies hence is less costly.

Disadvantages:

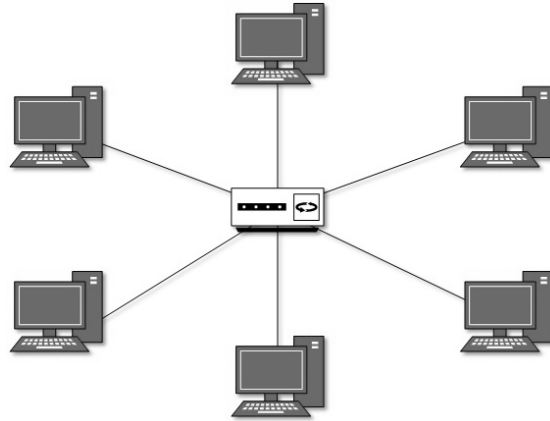
- Difficulty in reconnection and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices.
- Signal reflection at the taps can cause degradation in quality. This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable.
- a fault or break in the bus cable stops all transmission, even between devices on the same side of the problem.
- The damaged area reflects signals back in the direction of origin, creating noise in both directions.

Star Topology

In this topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not directly linked to one another. The controller acts as an exchange.

The hub device can be:

- ✓ Layer-1 device such as hub or repeater
- ✓ Layer-2 device such as switch or bridge
- ✓ Layer-3 device such as router or gateway



Like in Bus topology where communication line acts as a single point of failure, hub acts as single point of failure. If hub fails, connectivity of all hosts to all other hosts fails. Every communication between hosts takes place through only the hub.

Star topology is not expensive as to connect one more host, only one cable and one I/O port is needed to connect it to any number of others and configuration is simple.

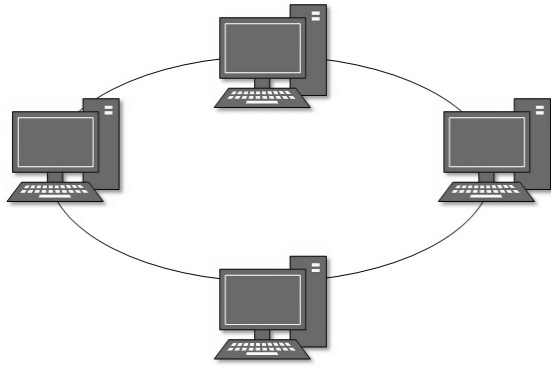
Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.

Other advantages include robustness. If one link fails, only that link is affected. This factor also lends to easy fault identification and fault isolation.

One **big disadvantage** of a star topology is the dependency of the whole topology on one single point, the hub.

Ring Topology

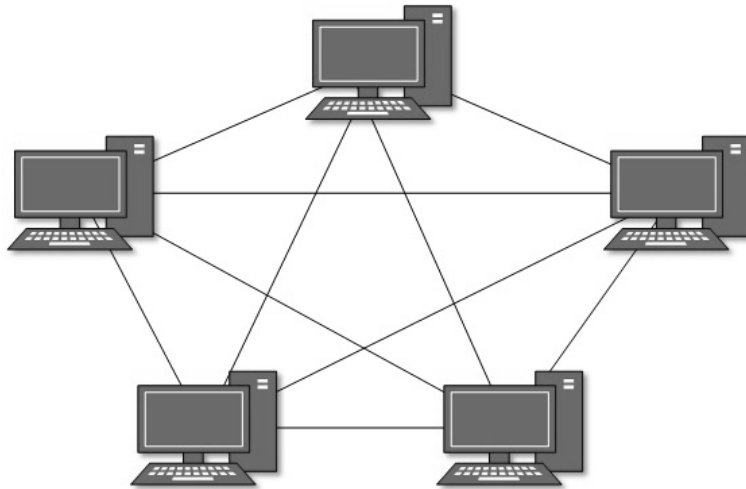
In ring topology, each host machine connects to exactly two other machines, creating a circular network structure. When one host tries to communicate or send message to a host which is not adjacent to it, the data travels through all intermediate hosts. To connect one more host in the existing structure, the administrator may need only one more extra cable.



Failure of any host results in failure of the whole ring. Thus, every connection in the ring is a point of failure. There are methods which employ one more backup ring.

Mesh Topology

In this topology, every device has a dedicated point-to-point link to every other device. The term **dedicated** means that the link carries traffic only between the two devices it connects.



Hosts in Mesh topology also work as relay for other hosts which do not have direct point-to-point links. Mesh technology comes into two types:

1. **Full Mesh:** All hosts have a point-to-point connection to every other host in the network. Thus for every new host $n(n-1)/2$ connections are required. It provides the most reliable network structure among all network topologies.
2. **Partially Mesh:** Not all hosts have point-to-point connection to every other host. Hosts connect to each other in some arbitrary fashion. This topology exists where we need to provide reliability to some hosts out of all.

Advantages:

1. The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
2. A mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
3. Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
4. Point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

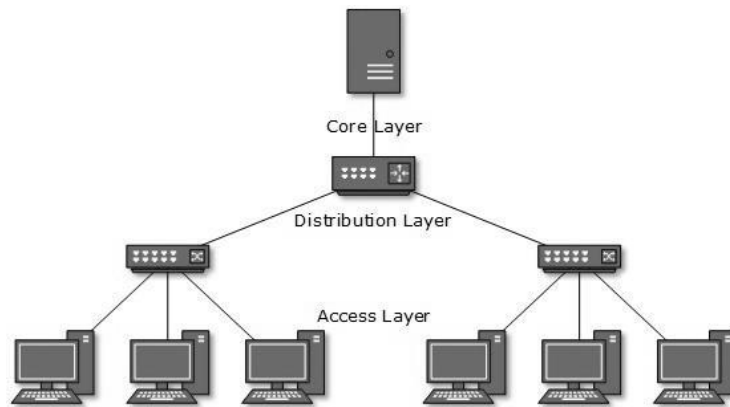
Disadvantages:

1. The amount of cabling required is high hence costly because every device must be connected to every other device, installation and reconnection are difficult.
2. The sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate. Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

Tree Topology

Also known as Hierarchical Topology, this is the most common form of network topology in use presently. This topology imitates an extended Star topology and inherits properties of Bus topology.

This topology divides the network into multiple levels/layers of network. Mainly in LANs, a network is bifurcated (divide into branches) into three types of network devices. The lowermost is access-layer where computers are attached. The middle layer is known as distribution layer, which works as mediator between upper layer and lower layer. The highest layer is known as core layer, and is central point of the network, i.e. root of the tree from which all nodes fork.



All neighboring hosts have point-to-point connection between them. Similar to the Bus topology, if the root goes down, then the entire network suffers even though it is not the single point of failure. Every connection serves as point of failure, failing of which divides the network into unreachable segment.

Daisy Chain

This topology connects all the hosts in a linear fashion. Similar to Ring topology, all hosts are connected to two hosts only, except the end hosts. Means, if the end hosts in daisy chain are connected then it represents Ring topology.

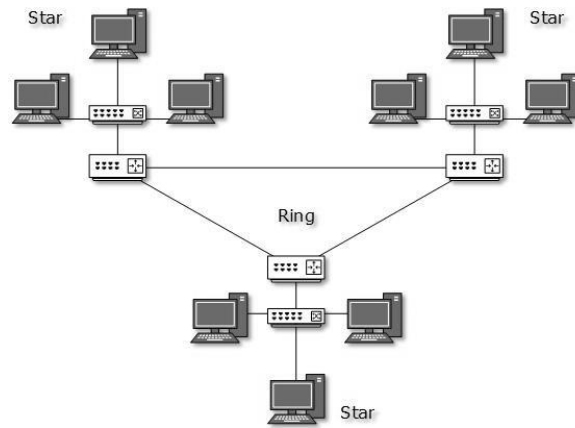
Each link in daisy chain topology represents single point of failure. Every link failure splits the network into two segments. Every intermediate host works as relay for its immediate hosts.



Hybrid Topology

A network structure whose design contains more than one topology is said to be hybrid topology. Hybrid topology inherits merits and demerits of all the incorporating topologies.

A network structure whose design contains more than one topology is said to be hybrid topology. Hybrid topology inherits merits and demerits of all the incorporating topologies.



The above picture represents an arbitrarily hybrid topology. The combining topologies may contain attributes of Star, Ring, Bus, and Daisy-chain topologies. Most WANs are connected by means of Dual-Ring topology and networks connected to them are mostly Star topology networks. Internet is the best example of largest Hybrid topology.